Richard Chiou

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Summary

Experienced AI/ML engineer with 10+ years designing, developing, and deploying production-grade machine learning and deep learning solutions across **computer vision, NLP, GenAI, and real-time analytics**. Led the **WildTrack project**, achieving 95%+ accuracy in species and individual footprint identification and LLM-based **Healthcare Virtual Assistant** which summarizes doctor-patient conversations and delivers personalized healthcare advice, improving clinical workflow efficiency by 60%. Developed an **AI-driven Traffic Intelligence System** that enhanced evacuation management and **supported the safe evacuation of residents during Hurricane Ian (2022).** Strong background in Python development, API integration (**Flask**, **FastAPI**), and cloud-based AI model deployment (**AWS**, **GCP**, **Azure**).

Hard Skills

- **Programming Languages:** Python, SQL
- **Frameworks & Libraries:** Django, Flask, FastAPI, Pandas, NumPy, Seaborn
- **ML Frameworks:** PyTorch, Tensorflow, Hugging Face, SageMaker, LangChain, LangGraph, Databricks
- **NLP & Computer Vision:** RAG, Fine-tunning, Transformer, CNN, RNN, LSTM, LLM, BERT, Chatbot, Speech-to-Text, Text-to-Speech, OpenCV, Graph WaveNet, Hadoop, **Model Context Protocol (MCP)**
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, PostGIS, Spark, Snowflake, Hive
- **DevOps & Cloud:** Docker, Jenkins, AWS (SageMaker, Lambda, EC2, EKS)
- **Simulation & Testing:** SUMO, A/B Testing, PyTest, UnitTest

WORK EXPERIENCE

Outlier | Newark, CA

Lead AI/ML Engineer | 04.2024 – Present

- Fine-tuned **computer vision and CNN models (ResNet-50)** using PyTorch to identify species, individual animals, sex, and age-class from wildlife footprint images, **achieving over 95% accuracy**.
- Integrated optimized ONNX models into **WildTrackAI mobile app, reducing prediction time from 2s to 300ms**, improving accessibility for **1000+ field users.**
- Implemented a multi-agent healthcare assistant using **Model Context Protocol (MCP)** to standardize tool access across patient data retrieval, summarization, and recommendation services, enabling modular and extensible agent workflows.
- Deployed a **BioBERT-based LLM** model using LangChain-powered RAG on SageMaker to **handle 500+ patient conversations per month**, providing personalized healthcare advice.
- Improved clinical workflow efficiency, **reducing average doctor-patient meeting time by 30%** by pre-summarizing patient information from virtual assistant conversations using **a BART-based Transformer model**.
- **Built and deployed scalable RESTful APIs using FastAPI and Flask** to serve LLM inference, conversation summarization, and healthcare recommendations, supporting real-time interactions with sub-second latency.
- Led a cross-functional AI team of 7 members to implement **reproducible MLOps workflows** using MLflow, DVC, Docker, and AWS S3, **reducing model deployment time by 40%** and enabling scalable, cloud-based AI operations.

Technologies: PyTorch, TensorFlow, ONNX, ResNet-50, OpenCV, LLM, RAG, BioBERT, BART, TTS/STT, LangChain, PostgreSQL, AWS (Lambda, S3, EC2), Databricks, SageMaker, MCP

StreetLight Data | San Francisco, CA

Founding AI/ML Engineer | 01.2022 – 04.2024

- Architected **spatio-temporal forecasting models using Graph WaveNet and LangGraph** to predict traffic flow one hour ahead, achieving 23% higher accuracy compared to ARIMA.
- Applied **Transformer-based temporal encoders** for multi-horizon congestion prediction across heterogeneous inputs, enabling early detection of evacuation bottlenecks and **reducing congestion prediction error by 18%.**
- Created **real-time incident detection models** combining loop detector data, GPS traces, and DOT event feeds to classify crashes and stalled vehicles **with 91% precision and 87% recall**, reducing operator detection latency by 35%.

- Built **a reinforcement learning model** for adaptive traffic signal timing and contraflow recommendations, validated in SUMO simulations with **up to 18% improvement in throughput** during peak evacuation load.
- Established real-time traffic AI pipelines on AWS, **ingesting 500K+ daily GPS and sensor records** through Kafka, preprocessing with Spark, and deploying models on SageMaker with **99.9% uptime for high availability.**
- Designed and implemented scalable data pipelines using **Snowflake** as the central data warehouse, ingesting and processing **2M–3M records per day**, enabling **1s query latency** and accelerating feature extraction for ML and LLM pipelines by **35%**.
- Built ELT workflows integrating Snowflake with AWS S3 and Databricks, reducing end-to-end data processing latency by **45%** and increasing data availability SLAs to **99.5%**, supporting reliable model training and near real-time inference.

Technologies: PyTorch, TensorFlow, scikit-learn, Graph WaveNet, Transformer, RL, Kafka, Spark, EKS, Redis, Prometheus, SUMO, PostGIS, WhyLabs, Hive, Airflow, Linux, LangGraph, Snowflake, ELT, S3, Databricks

Metis | San Francisco, CA

Data Scientist, Instructor | 02.2020 - 01.2022

- Mentored learners in feature engineering, model evaluation, and deployment, resulting in measurable skill improvements and career placements for **70+ students in data science roles.**
- Authored 50+ hands-on exercises and guided **20+ capstone projects**, producing production-ready ML models and analytics solutions.

Technologies: TensorFlow, PyTorch, Matplotlib, Pandas, Seaborn, Flask, FastAPI, Docker, Git/GitHub, AWS, GCP, Azure

Signifyd | San Jose, CA

AI/ML Engineer | 10.2018 – 02.2020

- Engineered machine learning models to **detect fraudulent e-commerce transactions**, processing 100K+ daily transactions and reducing chargebacks by 15%.
- **Optimized adaptive fraud thresholds** and conducted **A/B tests**, improving legitimate transaction approval rates by 10% while maintaining high fraud detection accuracy.

Technologies: XGBoost, scikit-learn, Pandas, NumPy, SQL, A/B tests, FastAPI, AWS

Brain Technologies | San Mateo, CA

Senior Data Scientist | 07.2017 - 09.2018

- Formulated data generation guidelines and managed data generation workflows, **creating and annotating 50,000+ examples** to support the NLP team and ensure high- quality training data.
- Operationalized general and domain-specific entity recognition systems using spaCy and SENNA, improving the virtual **assistant's entity extraction accuracy by 25%.**

Technologies: spaCy, NumPy, Pandas, NER, Stranford SENNA, NLP

xAd | Mountain View, CA

Mid-level Data Scientist | 07.2016 - 06.2017

- Executed **a patent-pending statistical algorithm** using hierarchical and Bayesian modeling on 10M observations to **improve ad campaign-driven store visits by 25%.**
- Developed a model to **rank mobile user id quality** leveraging 30+ behavioral and demographic features, **increasing targeting precision by 20%** across campaigns.

Technologies: Python, SQL, PyMC3, Matplotlib, Hierarchical Modeling, Hadoop

PayPal | San Jose, CA

Mobile Tools Development Intern | 01.2014 - 06.2016

- Constructed the **PPUtils tool using QIJaws API** to automate the creation of 10K+ test user accounts, supporting continuous integration for PayPal iOS and Android apps.

- Performed root cause analysis on Android test failures and **built a login performance dashboard** with automated alerts, improving testing efficiency by 45%.

Technologies: Python, QIJaws REST API, CI/CD, Watchtower, iOS & Android

Education

University of California | Berkeley, CA - Master's Degree, Electrical Engineering and Computer Science *08.2015 – 05.2016*

Columbia University | New York - Bachelor's Degree, Computer Science *09.2011 – 05.2015*